

HANZLA SIBGHAT

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WORK EXPERIENCE

Siemens AG

Software Engineer (Working Student)

Munich, Germany

Feb 2025 – Present

- Architected a full-stack enterprise Chrome extension with agentic AI capabilities (multi-step reasoning, tool use, and autonomous document workflows) for internal Siemens knowledge bases, improving document search efficiency by 35% across thousands of active users
- Engineered high-performance RAG pipelines and document ingestion flows integrating LangChain, GPT-4, and Mistral with vector embeddings, implementing VLM-based screenshot understanding and OCR-based processing to parse complex multi-modal PDFs with 95%+ extraction accuracy
- Designed a local IndexedDB-backed Pub/Sub architecture and Chrome extension message-passing broker for asynchronous job pipelines, reducing communication overhead and UI latency by 45%
- Developed rich, real-time user interfaces in React/TypeScript and Redux, achieving a 50% reduction in initial page load times through dynamic code-splitting and layout optimizations
- Streamlined deployment infrastructure by containerizing services using Docker and establishing automated CI/CD via GitHub Actions, cutting build-to-deploy cycles from 30 minutes to under 5 while maintaining 80%+ test coverage with Jest

National Database and Registration Authority (NADRA)

Software Engineer

Islamabad, Pakistan

Feb 2024 – Sept 2024

- Designed and scaled RESTful APIs handling 10k+ requests per minute utilizing Nginx load balancing, connection pooling, and Redis caching, which reduced average database query latency by 70% for national identity services
- Implemented event-driven architecture with Apache Kafka and RabbitMQ for inter-service communication, decoupling mission-critical citizen validation workflows and increasing overall system throughput
- Engineered citizen-facing CNIC application and renewal workflows (One App) with multi-step validation and Family Registration Certificate (FRC) integration, boosting application success rates by 30% and reducing processing times by 40%
- Integrated IoT biometric validation devices (fingerprint/iris scanners) with Java/Spring Boot backend and React kiosk interfaces, automating payment and validation flows to decrease center queue lengths by 30% and optimize operations by 45%
- Maintained strict security protocols by implementing TLS-secured communications and signed payloads for data transfer across distributed national database nodes
- Enforced robust testing standards using JUnit, Mockito, and Jest for unit and integration testing, ensuring zero-downtime releases for nationwide government infrastructure

PLC Group

Software Engineer

Ontario, Canada (Remote)

July 2023 – Feb 2024

- Architected responsive monitoring dashboards for remote hardware site management across Philippines telecom infrastructure, resulting in a 30% reduction in physical on-site maintenance visits
- Designed optimized GraphQL schemas with DataLoader batching to eliminate N+1 query issues, improving average API response latencies by 60% for 100k+ concurrent users
- Built real-time data visualizations and performance histograms using React, React Query, and Redux Toolkit, enhancing customer decision-making speed by 50%
- Established high-availability deployment pipelines utilizing blue-green deployment strategies and Redis caching layers, maintaining 100% platform uptime

SKILLS

Frontend: TypeScript, JavaScript, React, Next.js, Redux, HTML5, CSS3, Tailwind CSS, Material-UI

Backend & Infrastructure: Node.js, Express.js, Spring Boot, PostgreSQL, MongoDB, MySQL, Redis, Docker, AWS, GitHub Actions, Kafka, RabbitMQ, Nginx, Linux, Vercel

AI, Testing & Tools: RAG Architecture, LangChain, Vector Embeddings, Semantic Search, OpenAI API, Ollama, Jest, JUnit, Playwright, Cypress, Git, Postman

Languages: English (C1), German (A2), Urdu (Native)

EDUCATION

University of Passau

M.Sc. in Computer Science

Passau, Germany

Oct 2024 – Oct 2026

FAST National University of Computer and Emerging Sciences

B.Sc. in Software Engineering

Islamabad, Pakistan

Aug 2019 – Jun 2023